

# Guidelines for Preparing the Presentation Slides for Intro to Data Science and AI Project

This document provides the structure and expectations for students submitting a presentation for the project on **Exploratory Data Analysis (EDA), K-NN Classification, and K-Means Clustering**. Ensure your slides are concise, visually appealing, and convey your analysis and findings clearly.

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## Slide Structure and Content

### 1. Title Slide

- **Project Title:** **Predicting E-Commerce Customer Retention**
- **Student's Name:** Include your name
- **Submission Date:** Add the date of submission.

### 2. Introduction

- **Objective:** Summarize the project's aim in 2-3 concise bullet points.
- **Dataset Overview:** Briefly describe the dataset:
  - Number of records and features.
  - Key columns included.

- **Tools and Techniques:** List the tools/software used for analysis (e.g., Excel, Python, Google Sheets) and any methods or algorithms applied.

### 3. Exploratory Data Analysis (EDA)

- Present key findings from the dataset, such as:
  - Maximum or minimum values.
  - Averages, medians, or frequently occurring values.
  - Observed trends or patterns.
- **Visual Representation:** Use:
  - Graphs (e.g., bar charts, line graphs) to highlight trends.
  - Tables to summarize statistics concisely.

### 4. Methodology

- Explain the approach used for each part of the project:

#### (i) K-NN Classification:

- Describe the steps: distance calculation, selecting nearest neighbors, and predicting outcomes.

#### (ii) K-Means Clustering:

- Detail initialization, assignment, and recomputation steps.
- Include brief, clear flow diagrams or step-by-step visuals to make the process easy to follow.

## 5. Results

- **K-NN Classification Results:**
  - Show predicted outcomes with supporting evidence like distances.
- **K-Means Clustering Results:**
  - Highlight final cluster assignments and updated centroids using tables or graphs.
- Ensure results are clearly linked to the methodology.

## 6. Insights and Learnings

- Summarize key takeaways:
  - Trends or patterns discovered during the analysis.
  - Unique insights into the dataset.
- Highlight how the analysis tools or machine learning techniques contributed to solving the problem.

## 7. Challenges and Recommendations

- **Challenges:**
  - Mention any issues encountered, such as computational difficulties.
- **Recommendations:**
  - Suggest improvements for future projects

## 8. Conclusion

- Recap the project's purpose and its main findings.
- Discuss the broader implications of the results.
- Emphasize the value of data science and AI in addressing similar problems.

## 9. References

- List any resources used, including:
    - Tools and software.
    - Websites, articles, or books referenced.
  - Format references neatly and consistently.
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## Design Tips

- **Keep Slides Clean:** Avoid overcrowding; use bullet points instead of long paragraphs.
  - **Use Visuals:** Incorporate graphs, charts, and tables wherever possible.
  - **Consistent Formatting:** Use the same font, color scheme, and layout across all slides.
  - **Limit Text:** Use concise language; each slide should ideally not exceed 6 bullet points.
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## Submission Details

- **Format:** Submit the presentation as a PDF or PowerPoint file.
- **Deadline:** Ensure submission by the specified deadline.
- **File Naming Convention:**
  - Use the format: `Registered_mail_ID_ProjectTitle`
  - (e.g., `John@gmail.com_Predicting_E-Commerce_Customer_Retention`).